

Tutorial Book



B.Sc Drive Technology

D. T. McGuiness, Ph.D

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This TutorialBook is a
complement to it's corresponding Lecture.
When in doubt regarding theory, please consult the LectureBook.



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1

Magnetic Circuits and Materials

Q1 A Simple Magnetic Circuit

The magnetic circuit shown below has dimensions $A_c = 9 \times 10^{-4} \text{ m}^2$, $A_g = 9 \times 10^{-4} \text{ m}^2$, $g = 5 \times 10^{-4} \text{ m}$, $l_c = 0,3 \text{ m}$, and $N = 500$.

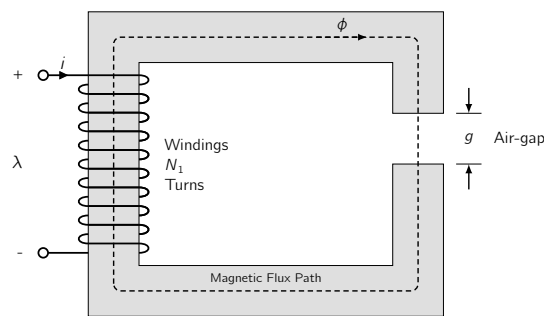


Figure 1.1: A simple magnetic circuit.

Assume the value $\mu_r = 70000$ for core material. Based on the information given above:

- Find the core ($\mathcal{R}_c$) and air-gap ($\mathcal{R}_g$) reluctance.

For the condition that the magnetic circuit is operating with $B_c = 1 \text{ T}$,

- Find the flux (ϕ) flowing through the circuit,
- Find the current (i) passing through the winding.

$$\text{(a)} \cdot 3,789,403 \times 10^3 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1}, \quad 4,420,970.64 \times 10^5 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1}, \quad \text{(b)} \cdot 9 \times 10^{-4} \text{ Wb}, \quad \text{(c)} \cdot 0,803 \text{ A}.$$

Answer

A1

- The reluctances can be found as follows:

$$\mathcal{R}_c = \frac{l_c}{\mu_r \mu_0 A_c} = \frac{0,3}{7,0000 \times 10^4 (4\pi \times 1 \times 10^{-7}) (9 \times 10^{-4})} = 3,789.403 \times 10^3 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1} \quad \blacksquare$$

$$\mathcal{R}_g = \frac{g}{\mu_0 A_g} = \frac{5 \times 10^{-4}}{(4\pi \times 1 \times 10^{-7}) (9 \times 10^{-4})} = 4,420.970.64 \times 10^5 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1} \quad \blacksquare$$

- The flux (ϕ) is found as follows:

$$\phi = B_c A_c = 1.0 (9 \times 10^{-4}) 9 \times 10^{-4} \text{ Wb} \quad \blacksquare$$

c. The current is found from the relation ($\mathcal{F} = Ni$):

$$i = \frac{\mathcal{F}}{N} = \frac{\phi(\mathcal{R}_c + \mathcal{R}_g)}{N} = \frac{9 \times 10^{-4} (4,46 \times 10^5)}{500} = 0,803 \text{ A} \quad \blacksquare$$

Q2

A Simple Machine

The magnetic structure of a synchronous machine is shown schematically below. Assuming both rotor and stator have infinite permeability ($\mu \rightarrow \infty$), find the air-gap flux (ϕ_g) and flux density B_g . For this example: assume the values $I = 10 \text{ A}$, $N = 1,000 \times 10^3 \text{ Turn}$, $g = 0,01 \text{ m}$, and $A_g = 0,02 \text{ m}^2$.

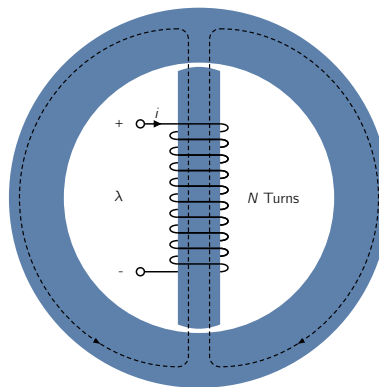


Figure 1.2: A simple synchronous machine, the air gap has a length of g and both the stator and rotor have infinite permeability.

$$B_g = 0,6285 \text{ T}$$

Answer

A2

Notice there are two air gaps in series, of total length $2g$, and by symmetry the flux density in each is **equal**. Given the iron permeability is infinite, its reluctance is assumed negligible and Eq. (1.20), where g is replaced by the total gap length $2g$ can be used to find the flux:

$$\phi = \frac{NI\mu_0 A_g}{2g} = \frac{(1,000 \times 10^3) (10) (4\pi \times 10^{-7}) (0,02)}{2 \times 0,01} = 0,012,57 \text{ Wb}$$

and

$$B_g = \frac{\phi}{A_g} = \frac{0,012,57}{0,02} = 0,6285 \text{ T}$$

Q3

Analysing a Magnetic Circuit

A magnetic circuit consists of an N -turn winding on a magnetic core of **infinite** permeability with two **parallel** air gaps of lengths g_1 and g_2 and areas A_1 and A_2 , respectively. Using this information, please calculate:

- The inductance of the winding (L),
- The flux density B_1 in gap 1 when the winding is carrying a current i .

Note: Neglect fringing effects at the air gap.

$$\frac{\tau_g}{N\mu_0 l} = \frac{\tau_A}{\tau_A} = \tau_B \cdot (\mathbf{q}) \quad \left(\frac{z_g}{A} + \frac{\tau_g}{\tau_A} \right) z N^0 l = \tau \cdot (\mathbf{e})$$

Answer

A3

- a. Once drawn, the equivalent circuit shows the total reluctance is equal to the parallel combination of the two (2) gap reluctances. Therefore:

$$\phi = \frac{Ni}{\frac{\mathcal{R}_1 \mathcal{R}_2}{\mathcal{R}_1 + \mathcal{R}_2}}$$

where:

$$\mathcal{R}_1 = \frac{g_1}{\mu_0 A_1} \quad \text{and} \quad \mathcal{R}_2 = \frac{g_2}{\mu_0 A_2}.$$

Using the relation $\lambda = Li$, we get the following:

$$L = \frac{\lambda}{i} = \frac{N\phi}{i} = \frac{N^2 (\mathcal{R}_1 + \mathcal{R}_2)}{\mathcal{R}_1 \cdot \mathcal{R}_2} = \mu_0 N^2 \left(\frac{A_1}{g_1} + \frac{A_2}{g_2} \right) \quad \blacksquare$$

- b. From the magnetic equivalent circuit, one can see that

$$\phi_1 = \frac{\mathcal{F}}{\mathcal{R}_1} = \frac{Ni}{\mathcal{R}_1} = \frac{\mu_0 A_1 Ni}{g_1} \quad \text{therefore} \quad B_1 = \frac{\phi_1}{A_1} = \frac{\mu_0 Ni}{g_1} \quad \blacksquare$$

Q4

Magnetic Circuit with Core Permeability

In **Ex. 1**, the relative permeability of the core material for the magnetic circuit below is assumed to be $\mu_r = 7,0000 \times 10^4$ at a flux density of 1 T.

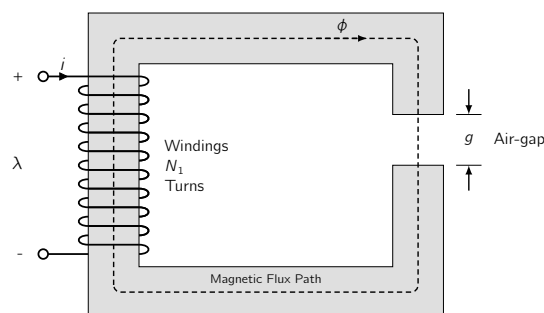


Figure 1.3: The magnetic circuit in question.

- a. In a practical device, the core would be constructed from electrical steel such as M-5 electrical steel. This material is highly non-linear and its relative permeability varies from a value of approximately $\mu_r = 7,2300 \times 10^4$ at a flux density of $B = 1$ T to a value of on the order of $\mu_r = 2,900 \times 10^3$ as the flux density is raised to 1,8 T. Based on this information, please calculate the inductance under the assumption that the relative permeability of the core steel is $7,2300 \times 10^4$.

- b. Calculate the inductance under the assumption that the relative permeability (μ_r) is equal to $2,900 \times 10^3$.

$$0.468 \text{ H} \quad (a) \quad 0.561 \text{ H} \quad (b)$$

Answer

A4

- a. Based upon the dimensions given in **Ex. 1**:

$$\mathcal{R}_c = \frac{l_c}{\mu_r \mu_0 A_c} = \frac{0.3}{7,2300 \times 10^4 (4\pi \times 10^{-7}) (9 \times 10^{-4})} = 3,67 \times 10^3 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1}$$

while $\mathcal{R}_g$ remains unchanged from the value calculated in **Ex. 1** as $\mathcal{R}_g = 4,42 \times 10^5 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1}$. Therefore the total reluctance of the core and gap is

$$\mathcal{R}_{\text{tot}} = \mathcal{R}_c + \mathcal{R}_g = 4,46 \times 10^5 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1}$$

and therefore:

$$L = \frac{N^2}{\mathcal{R}_{\text{tot}}} = \frac{500^2}{4,46 \times 10^5} = 0,561 \text{ H} \quad \blacksquare$$

- b. For $\mu_r = 2,900 \times 10^3$, the reluctance of the core ($\mathcal{R}_c$) increases from a value of:

$$\mathcal{R}_c = 3,79 \times 10^3 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1},$$

to a value of,

$$\mathcal{R}_c = \frac{l_c}{\mu_r \mu_0 A_c} = \frac{0.3}{2,900 \times 10^3 (4\pi \times 10^{-7}) (9 \times 10^{-4})} = 9,15 \times 10^4 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1},$$

and therefore the total reluctance increases from $4,46 \times 10^5 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1}$ to $5,34 \times 10^5 \text{ A} \cdot \text{Turn} \cdot \text{Wb}^{-1}$. Therefore the inductance decreases from 0,561 H to

$$L = \frac{N^2}{\mathcal{R}_{\text{tot}}} = \frac{500^2}{5,34 \times 10^5} = 0,468 \text{ H} \quad \blacksquare$$

Note: This illustrates the linearising effect of a dominating air gap in a magnetic circuit. In spite of a reduction in the permeability of the iron by a factor of $7,2300 \times 10^4 / 2,900 \times 10^3 = 25$, the inductance decreases only by a factor of $0,468 / 0,561 = 0.83$ simply because the reluctance of the air gap is significantly larger than that of the core. In many situations, it is common to assume the inductance to be constant at a value corresponding to a finite, constant value of core permeability (or in many cases it is assumed simply that $\mu_r \rightarrow \infty$). Analyses based upon such a representation for the inductor will often lead to results which are well within the range of acceptable engineering accuracy and which avoid the immense complication associated with modeling the non-linearity of the core material.

Q5

Plotting Permeability

Using Python, plot the inductance of the magnetic circuit given in **Ex. 1** as a function of core permeability over the range $100 \leq \mu_r \leq 100000$.

A5

The script for the plot is as follows:

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  mu0 = 4 * np.pi * 1e-7      # (H/m) permeability of free space
5  lc  = 0.3                    # (m) length
6  Ac  = 9e-4                   # (m2) cross-sectional area
7  Ag  = 9e-4                   # (m2) cross-sectional area
8  g   = 5e-4                   # (m) air gap length
9  N   = 500                    # (-) number of turns
10  Bc  = 1                      # (T) Core magnetic field density
11
12  Rel_g = (g) / (mu0 * Ag)     # (Ampere-Turns/Weber) air-gap reluctance
13
14  mur = np.zeros(1000)        # Relative Permeability empty array for looping
15
16  for i in range(len(mur)):
17      mur[i] = 100 + (100000 - 100) * (i - 1)/1000
18
19
20  Rel_c = (lc) / (mur * mu0 * Ac) # (Ampere-Turns/Weber) core reluctance
21  R_tot = Rel_c + Rel_g          # (Ampere-Turns/Weber) total reluctance
22
23  L = N**2 / R_tot              # (H) Inductance
24
25  # -- Plotting
26  plt.xlabel('Core Relative Permeability ($\mu_r$) [H/m]')
27  plt.ylabel('Inductance [H]')
28  plt.xlim(0, 0.8e5)
29  plt.ylim(0, 1)
30  plt.title("Core Relative Permeability vs. Inductance Graph", color='#c6d0f5')
31  plt.plot(mur, L, '#ef9f76', linewidth=5)
32  fig.tight_layout()
33  plt.savefig('images/python-matplot-fig.svg')

```

Please observe the resultant plot. The figure clearly confirms that, for the magnetic circuit of this example, the inductance is quite **insensitive** to relative permeability until the relative permeability drops to on the order of 1000. Therefore, as long as the effective relative permeability of the core is “large”, which in this case greater than 1000, any non-linearities in the properties of the core material will have little effect on the terminal properties of the inductor.

Q6

Time Varying Voltage in Magnetic Circuit

For the magnetic circuit of **Ex. 1** with the following diagram:

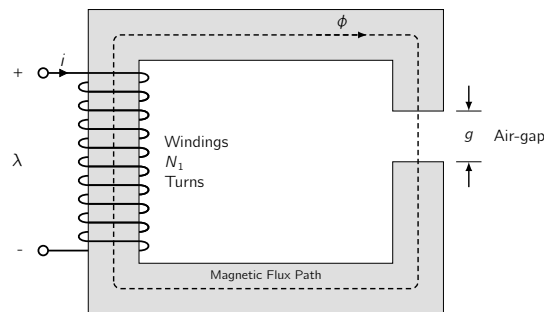


Figure 1.4: The circuit in question.

Find:

1. The inductance (L),
2. The magnetic stored energy (W) for $B_c = 1$ T.
3. The induced voltage (e) for a 60 Hz time-varying core flux of the form $B_c = 1 \sin \omega t$.

$$((1773) \cos(377t) \times 10^{-3}) \cdot (c) \quad '1773 \cdot 0.01 \cdot (q) \quad 'H 89 \cdot 095 \cdot 0 \cdot (e)$$

Answer

A6

- a. From the flux linkage relationship, we know that:

$$L = \frac{\lambda}{i} = \frac{N\phi}{i} = \frac{N^2}{\mathcal{R}_c + \mathcal{R}_g} = \frac{500^2}{4.46 \times 10^5} = 0.56 \text{ H}$$

Please observe the core reluctance is much smaller than that of the gap ($\mathcal{R}_c \ll \mathcal{R}_g$). Therefore, to a good approximation the inductance is dominated by the gap reluctance, i.e.,

$$L \approx \frac{N^2}{\mathcal{R}} = 0.57 \text{ H}$$

- b. In **Ex. 1** we found that when $B_c = 1$, T, $i = 0.80$ A. Therefore we can see the stored magnetic energy as,

$$W = \frac{1}{2} Li^2 = \frac{1}{2} (0.56)(0.80)^2 = 0.18 \text{ J}$$

- c. The induced voltage (e) is calculated as follows:

$$\begin{aligned} e &= \frac{d\lambda}{dt} = N \frac{d\phi}{dt} = N A_c \frac{dB_c}{dt} = 500 \times (0.0009) \times (377 \times 1.0 \cos(377t)) \\ &= 170 \cos(377t) \quad \blacksquare \end{aligned}$$

Transformers

Q7 The Core Loss of a Transformer

The core loss and exciting voltamperes for the core of a transformer at $B_{\max} = 1.5 \text{ T}$ and 60 Hz were found to be:

$$P_c = 16 \text{ W} \quad \text{and} \quad (VI)_{\text{rms}} = 20 \text{ VA}$$

and the induced voltage was: $274/\sqrt{2} = 194 \text{ V rms}$ when the winding has 200 turns. Using this information, please find the power factor (ϕ), the core-loss current (I_c), and the magnetizing current (I_m).

$$\theta_c = -36.9^\circ, \quad \phi = 0.1 \text{ A}, \quad I_c = 0.082 \text{ A}, \quad I_m = 0.06 \text{ A}$$

Answer

A7

Power factor is:

$$\cos \theta_c = \frac{16}{20} = 0.80 \quad (\text{lag}), \quad \text{therefore} \quad \theta_c = -36.9^\circ$$

Note that we know that the power factor is lagging because the system is **inductive**. The other values are:

$$\begin{aligned} \text{Exciting Current} \quad I_\phi &= \frac{20}{194} = 0.10 \text{ A rms} \\ \text{Core-loss Component} \quad I_c &= \frac{16}{194} = 0.082 \\ \text{Magnetising Component} \quad I_m &= I_\phi |\sin \theta_c| = 0.060 \text{ A rms} \quad \blacksquare \end{aligned}$$

Q8 The Equivalent Circuit of a Transformer

The equivalent circuit of shows an ideal transformer with an impedance

$$R_2 + jX_2 = 1 + j4 \Omega$$

connected in **series** with the secondary. The turns ratio is given as $(N_1/N_2) = 5 : 1$.

- Draw an equivalent circuit with the series impedance referred to the primary side.
- For a primary voltage of 120 V RMS and a short connected across the terminals a-b, calculate the primary current and the current flowing in the short.

A8

- a. The new equivalent is shown below. The secondary impedance is referred to the primary by the turns ratio squared. Therefore:

$$R'_2 + jX'_2 = \left(\frac{N_1}{N_2}\right)^2 (R_2 + jX_2) = 25 + j100\Omega \quad \blacksquare$$

- b. From Eq. (2.29), a short at terminals A-B will appear as a short at the primary of the ideal transformer, since the zero voltage of the short is reflected by the turns ratio N_1/N_2 to the primary.

Therefore the primary current will be given by:

$$\hat{I}_1 = \frac{\hat{V}_1}{R'_2 + jX'_2} = \frac{120}{25 + j100} = 0.28 - j1.13 \text{ A RMS}$$

corresponding to a magnitude of 1.16 A RMS. From Eq. (2.13), the secondary current will equal $N_1/N_2 = 5$ times that of the current in the primary. Thus the current in the short will have a magnitude of $5 \times (1.16) = 5.8 \text{ A RMS}$ ■

Q9

Programming Analysis of a Transformer

A 2400 :120-V, 50 Hz potential transformer has the following parameter values (referred to the 2400 V winding):

$$\begin{aligned} X_1 &= 143\Omega & X'_2 &= 164\Omega & X_m &= 1,630,000 \times 10^5\Omega \\ R_1 &= 128\Omega & R'_2 &= 141\Omega. \end{aligned}$$

Based on this information, assuming a 2400 V input, which ideally should produce a voltage of 120 V at the low-voltage winding, calculate the magnitude and relative phase-angle errors of the secondary voltage if the secondary winding is open-circuited.

A9

This problem can easily be solved using a Python script. From Eq. (2.35) with $V_1 = 2400 \text{ V}$, the result should give:

$$119,895 \angle 0.045^\circ \text{ V} \quad \blacksquare$$

Q10

Multi-Single-Phase Transformers

Three single-phase, 50-kVA 2400:240-V transformers, are connected Y- Δ in a three-phase 150-kVA bank to step down the voltage at the load end of a feeder whose impedance is $0.15 + j1.00\Omega/\text{phase}$. The voltage at the sending end of the feeder is 4160 V line-to-line. On their secondary sides, the transformers supply a balanced three-phase load through a feeder whose impedance is $0.0005 + j0.0020\Omega/\text{phase}$.

Find the line-to-line voltage at the load when the load draws rated current from the transformers at a power factor of 0.80 lagging.

A10

The calculations can be made on a single-phase basis by referring everything to the high-voltage, Y-connected side of the transformer bank. The voltage at the sending end of the feeder is equivalent to a source voltage V_s of:

$$V_s = \frac{4160}{\sqrt{3}} = 2,400 \times 10^3 \text{ V line-to-neutral}$$

From the transformer rating, the rated current on the high-voltage side is $20,8 \text{ A} \cdot \text{phase}^{-1}$ Y-connected. The low-voltage feeder impedance referred to the high voltage side by means of the square of the rated line-to-line voltage ratio of the transformer bank is:

$$Z_{lv,H} = \left(\frac{4160}{240} \right)^2 (0.0005 + j0.0020) = 0.15 + j0.60 \Omega$$

and the combined series impedance of the high- and low-voltage feeders referred to the highvoltage side is thus:

$$Z_{\text{feeder},H} = 0.30 + j1.60 \Omega / \text{phase Y}$$

Because the transformer bank is Y-connected on its high-voltage side, its equivalent single-phase series impedance is equal to the single-phase series impedance of each single-phase transformer as referred to its high-voltage side. This impedance is calculated as:

$$Z_{eq,H} = 1.42 + j1.82 / \text{phase Y}$$

The actual line-neutral load voltage can then be calculated by referring this value to the low-voltage side of the transformer bank as

$$V_{\text{load}} = 2329 \left(\frac{240}{4160} \right) = 134 \text{ V line-to-neutral}$$

which can be expressed as a line-to-line voltage by multiplying by $\sqrt{3}$

$$V_{\text{load}} = 134\sqrt{3} = 233 \text{ V line-to-line} \quad \blacksquare$$

Q11

A Poly-phase Transformer

Three standard transformers are reconnected Δ - Δ and supplied with power through a 2400-V (line-to-line) three-phase feeder whose reactance is $0,80 \Omega \cdot \text{phase}^{-1}$.

At its sending end, the feeder is connected to the secondary terminals of a three-phase Y – Δ -connected transformer whose rating is:

$$500 \text{ kV} \cdot \text{A}, \quad 24 \text{ kV} : 2,400 \times 10^3 \text{ V line-to-line}$$

The equivalent series impedance of the sending-end transformer is:

$$0.17 + j0.92 \Omega \cdot \text{phase}^{-1}$$

referred to the 2400-V side. The voltage applied to the primary terminals of the sending-end transformer is 24.0 kV line-to-line.

A three-phase short circuit occurs at the 240-V terminals of the receiving-end transformers.

Using this information, calculate the steady-state short-circuit current in the 2400-V feeder phase wires, in the primary and secondary windings of the receiving-end transformers, and at the 240-V terminals.

A11

The calculations will be made on an equivalent line-to-neutral basis with all quantities referred to the 2400-V feeder. The source voltage then is:

$$\frac{2400}{\sqrt{3}} = 1385V \quad \text{line-to-neutral}$$

From Eq. (2.35), the single-phase-equivalent series impedance of the Δ - Δ transformer seen at its 2400-V side is

$$Z_{eq} = R_{eq} + jX_{eq} = \frac{1.42 + j1.82}{3} = 0.47 + j0.61 \Omega \cdot \text{phase}^{-1}$$

The total series impedance to the short circuit is then the sum of this impedance, that of sending-end transformer and the reactance of the feeder

$$Z_{\text{tot}} = (0.47 + j0.61) + (0.17 + j0.92) + j0.80 = 0.64 + j2.33 \Omega \cdot \text{phase}^{-1}$$

which has a magnitude of

$$|Z_{\text{tot}}| = 2.42 \Omega \cdot \text{phase}^{-1}$$

The magnitude of the phase current in the 2400-V feeder can now simply be calculated as the line-neutral voltage divided by the series impedance

$$\text{Current in 2400-V feeder} = \frac{1385}{2.42} = 572A$$

and, the winding current in the 2400-V winding of the receiving-end transformer is equal to the phase current divided by $\sqrt{3}$ or

$$\text{Current in 2400-V windings} = \frac{572}{\sqrt{3}} = 330A$$

while the current in the 240-V windings is 10 times this value

$$\text{Current in 240-V windings} = 10 \times 330 = 3300A$$

Finally, the phase current at the 240-V terminals into the short circuit is given by

$$\text{Current at the 240-V terminals} = 3300\sqrt{3} = 5720A$$

Note of course that this same result could have been computed simply by recognizing that the turns ratio of the Δ - Δ transformer bank is equal to 10:1 and hence, under balanced-three-phase conditions, the phase current on the low voltage side will be 10 times that on the high-voltage side. ■

Q12

Voltage Regulation of a Transformer

With the instruments located on the high-voltage side and with the low-voltage side short-circuited, the short-circuit test readings for the 50-kVA 2400:240-V transformer of are 48 V, 20.8 A, and 617 W. An open-circuit test with the low-voltage side energised gives instrument readings on that side of 240 V, 5.41 A, and 186 W.

Determine the efficiency and the voltage regulation of the transformer operating at full load, 0.80 power factor **lagging**.

A12

From the short-circuit test, the magnitude of the equivalent impedance, the equivalent resistance, and the equivalent reactance of the transformer, here referred to the high-voltage side as denoted by the subscript H , are

$$\begin{aligned} |Z_{eq,H}| &= \frac{48}{20.8} = 2.31 \Omega & R_{eq,H} &= \frac{617}{20.82} = 1.42 \Omega, \\ X_{eq,H} &= \sqrt{2.31^2 - 1.42^2} = 1.82 \Omega. \end{aligned}$$

Operation at full-load, 0.80 power factor lagging corresponds to a current of

$$I_H = \frac{50000}{2400} = 20.8 \text{ A}$$

and an output power:

$$P_{\text{output}} = P_{\text{load}} = (0.8) 5,0000 \times 10^4 = 4,0000 \times 10^4 \text{ W}$$

Please observe that the short-circuit test was conducted at rated current and hence the full-load I^2R loss will equal that of the short-circuit test. Similarly, the open-circuit test was conducted at rated voltage and hence the full-load core loss is equal to that of the open-circuit test. As a result, the total loss under this operating condition is equal to the sum of the winding loss

$$P_{\text{winding}} = I_H^2 R_{eq,H} = 20.8^2 (1.42) = 617 \text{ W},$$

and the open-circuit core loss:

$$P_{\text{core}} = 186 \text{ W}$$

Therefore:

$$P_{\text{loss}} = P_{\text{winding}} + P_{\text{core}} = 803 \text{ W},$$

and the power input to the transformer is:

$$P_{\text{input}} = P_{\text{output}} + P_{\text{loss}} = 4,0803 \times 10^4 \text{ W}$$

The efficiency of a power conversion device is defined as:

$$\text{efficiency} = \frac{P_{\text{output}}}{P_{\text{input}}} = \frac{P_{\text{input}} - P_{\text{loss}}}{P_{\text{input}}} = 1 - \frac{P_{\text{loss}}}{P_{\text{input}}}$$

which can be expressed in percent by multiplying by 100 percent. Therefore, for this operating condition:

$$100\% \left(\frac{P_{\text{output}}}{P_{\text{input}}} \right) = 100\% \left(\frac{4,0000 \times 10^4}{4,0000 \times 10^4 + 803} \right) = 98.0\%$$

The voltage regulation of a transformer is defined as the change in secondary terminal voltage from no load to full load and is usually expressed as a percentage of the full-load value. In power systems applications, regulation is one figure of merit for a transformer; a low value indicates that load variations on the secondary of that transformer will not significantly affect the magnitude of the voltage being supplied to the load. It is calculated under the assumption that the primary voltage remains constant as the load is removed from the transformer secondary.

The primary voltage is assumed to be adjusted so that the secondary terminal voltage has its rated value at full load, or $V_{2H} = 2,400 \times 10^3 \text{ V}$. For a load of rated value and 0.8 power factor lagging, the load current will be

$$\hat{I}_H = \left(\frac{5,0 \times 10^4}{2,400 \times 10^3} \right) \exp(-j36,9^\circ) = 20,8 \exp(-j36,9^\circ) = 16,6 - j12,5 \text{ A}$$

The required value of the primary voltage V_H can be calculated as:

$$\begin{aligned} \hat{V}_{1H} &= \hat{V}_{2H} + \hat{I}_H (R_{eq,H} + jX_{eq,H}) \\ &= 2400 + (16,6 - j12,5) (1,42 + j1,82) \\ &= 2446 \exp(j0,29^\circ) \text{ V} \end{aligned}$$

The magnitude of $\hat{V}_{1H}$ is $2,446 \times 10^3 \text{ V}$. If this voltage were held constant and the load removed, the secondary voltage on open circuit would rise to $2,446 \times 10^3 \text{ V}$ referred to the high-voltage side. Therefore

$$\text{Regulation} = \left(\frac{2446 - 2400}{2400} \right) \times 100\% = 1,92\% \quad \blacksquare$$

3

Electromechanic Energy Conversion

Q13

Rotor Under a Field

A non-magnetic cylindrical rotor, which is mounted on a shaft at its centre, contains a single-turn coil. This rotor is placed in a **uniform magnetic field** of magnitude B_0 , as shown in **Fig. 3.1**. The coil sides are at radius R and the wire carries current I as indicated.

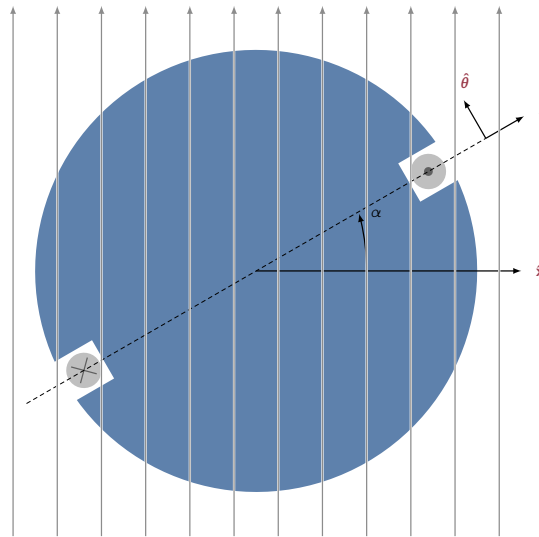


Figure 3.1: A non-magnetic cylindrical rotor, containing a single-turn coil. The rotor is under a uniform magnetic field (B_0).

Based on this information, find the θ -directed torque as a function of rotor position α when $I = 10 \text{ A}$, $B_0 = 0,02 \text{ T}$ and $R = 0,05 \text{ m}$.

Note: Assume that the rotor is of length $l = 0,3 \text{ m}$.

A13

The net current I in a wire is equal to the integral of the current density J over the cross-sectional area of the wire.

$$I = \int_{\text{wire}} J \cdot d\mathbf{A}.$$

Similarly, the net force per unit length on a wire in a uniform magnetic field B can be found from integrating Eq. (3.6) over the cross sectional area of the wire. Therefore:

$$\mathbf{F} = \left(\int_{\text{wire}} J \cdot d\mathbf{A} \right) \times \mathbf{B} = I \times \mathbf{B}$$

Using this information, for wire (call it wire 1) carrying current I **into** the paper, the θ -directed force is given by

$$F_{1\theta} = -IB_0 l \sin \alpha,$$

and for the wire carrying current in the opposite direction (call it wire 2) and is located 180° away from wire 1, the force is:

$$F_{2\theta} = -IB_0 l \sin \alpha,$$

where l is the length of the rotor. The torque T acting on the rotor is given by the sum of the force-moment-arm products for each wire:

$$\begin{aligned} T &= -2IB_0 R l \sin \alpha \\ &= -2(10)(0,02)(0,05)(0,3) \sin \alpha \\ &= -6 \times 10^{-3} \sin \alpha \quad \text{N} \cdot \text{m} \quad \blacksquare \end{aligned}$$

Q14 The Energy Stored in a Relay

The relay shown in **Fig. 3.4** is made from **infinitely-permeable** magnetic material with a movable plunger, which is also of infinitely-permeable material. The height of the plunger is much greater than the air-gap length ($h \gg g$).

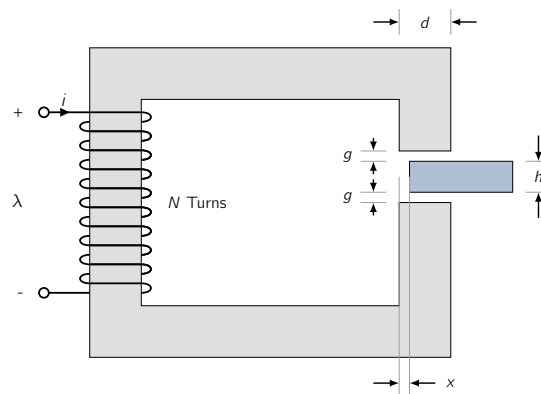


Figure 3.2: Relay with movable plunger. The relay has a depth of l .

Using this information, please calculate the magnetic stored energy W_{fld} as a function of plunger position ($0 < x < d$) for $N = 1000$ turns, $g = 2.0$ mm, $d = 0.15$ m, $l = 0.1$ m, and $i = 10$ A.

A14

Eq. (3.19) can be used to solve for W_{fld} when λ is known. For this situation, i is held **constant**, and therefore it would be useful to have an expression for W_{fld} as a function of both i and x .

This can be obtained quite simply by substituting Eq. (3.14) into Eq. (3.19), with the result

$$W_{\text{fld}} = \frac{1}{2} L(x) i^2$$

where the inductance is given by:

$$L(x) = \frac{\mu_0 N^2 A_{\text{gap}}}{2g}$$

where A_{gap} is the gap cross-sectional area (m^2). From **Fig. 3.4**, A_{gap} can be seen to be:

$$A_{\text{gap}} = l(d - x) = ld \left(1 - \frac{x}{d}\right)$$

Which means:

$$L(x) = \frac{\mu_0 N^2 l d \left(1 - \frac{x}{d}\right)}{2g}$$

and

$$\begin{aligned} W_{\text{fld}} &= \frac{1}{2} \frac{N^2 \mu_0 l d (1 - x/d)}{2g} i^2 \\ &= \frac{1}{2} \frac{(1000^2)(4\pi \times 10^{-7})(0.1)(0.15)}{2(0.002)} \times 10^2 \left(1 - \frac{x}{d}\right) \\ &= 236 \left(1 - \frac{x}{d}\right) \quad \text{J} \quad \blacksquare \end{aligned}$$

Q15

Egg-Shaped Rotor

The magnetic circuit of **Fig. 3.3** consists of a single-coil stator and an oval rotor.

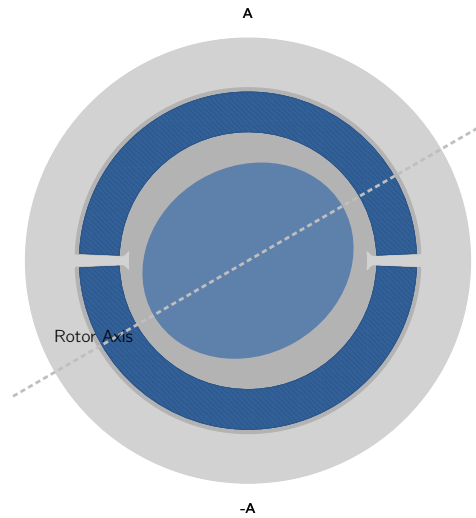


Figure 3.3: A Stator with an oval-shaped rotor.

Due to the air-gap being non-uniform, the coil inductance varies with rotor angular position, measured between the magnetic axis of the stator coil and the major axis of the rotor, as:

$$L(\theta) = L_0 + L_2 \cos(2\theta) \text{ H}$$

Using this information, please calculate the generated torque T_{fld} .

A15

From Eq. (3.33) we see that:

$$T_{\text{fld}}(\theta) = \frac{i^2}{2} \frac{dL(\theta)}{d\theta} = -L_2 i^2 \sin(2\theta) \text{ N} \cdot \text{m}$$

Substitution of given values to the equation gives:

$$T_{\text{fld}}(\theta) = -0.0108 \sin(2\theta) \text{ N} \cdot \text{m}$$

Note that in this case the torque acts in such a direction as to pull the rotor axis in alignment with the coil axis and hence to maximize the coil inductance ■

Q16 Determining from Coenergy

Relooking at the same relay again,

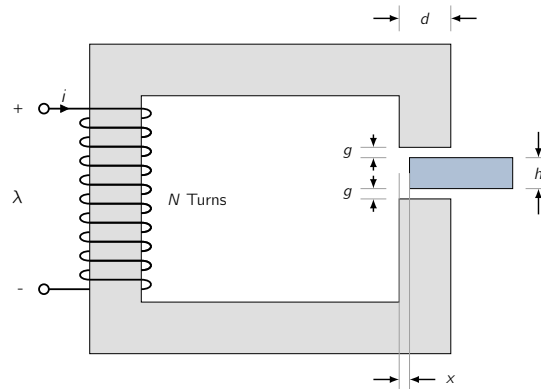


Figure 3.4: Relay with movable plunger. The relay has a depth of l .

Please find the force on the plunger as a function of x when the coil is driven by a controller which produces a current as a function of x of the form:

$$i(x) = \frac{l_0 x}{d} \text{ A}$$

A16

From **Ex. 14**, we have:

$$L(x) = \frac{N^2 d l \mu_0 (1 - \frac{x}{d})}{2g} \text{ H}$$

This is a magnetically-linear system for which the force can be calculated from Eq. (3.43) as:

$$f_{fld} = \frac{i^2}{2} \frac{dL(x)}{dx} = -\frac{N^2 i^2 l \mu_0}{4g} \text{ N}$$

Substituting for $i(x)$, the expression for the force as a function of x can be determined as:

$$f_{fld} = -\frac{l_0^2 N^2 l \mu_0 x^2}{4d^2 g} \text{ N}$$

Note that from Eq. (3.42), the coenergy for this system is equal to

$$W'_{fld}(i, x) = \frac{N^2 d i^2 l \mu_0 (1 - \frac{x}{d})}{4g} \text{ J}$$

and the derivative of this expression with respect to x gives the expected expression for the force in terms of the current i . In this example, one might be tempted to introduce the expression for $i(x)$ directly into the coenergy expression, in which case the coenergy would be given by:

$$W'_{fld}(i, x) = \frac{l_0^2 N^2 \mu_0 l d (1 - x/d)}{4\sigma} \left(\frac{x}{d} \right)^2$$

Although this is a perfectly correct expression for the coenergy as a function of x under the specified operating conditions, if one were to attempt to calculate the force from taking the partial derivative

of this expression for W'_{fld} with respect to x , the resultant expression would not give the correct expression for the force. The reason for this is quite simple: As seen from Eq. (3.40), the partial derivative must be taken holding the current constant. Having substituted the expression for $i(x)$ to obtain this equation for the coenergy, the current is no longer a constant and this requirement cannot be met. This illustrates the problems that can arise if the various force and torque expressions developed here are misapplied.

Q17 Multiply Excited System

Consider the following system with multiple excitation. Here, the magnetic circuit (i.e., stator is excited by a voltage source, and the rotor is excited from another,

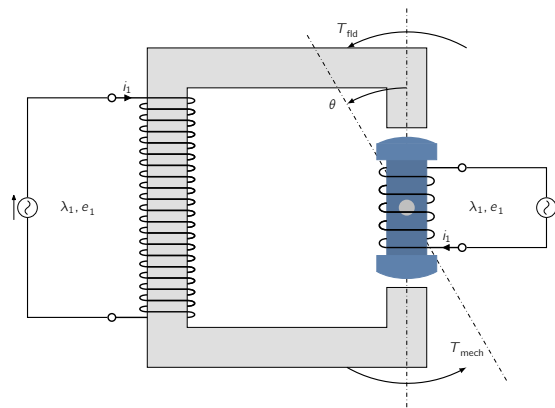


Figure 3.5: A system with multiple excitation. Here, the rotor is connected to a spring, which is then connected to a wall for stabilisation.

The inductances in henrys are given as:

$$L_{11} = 0.001 \cos(2\theta) + 0.003 \text{ H}$$

$$L_{12} = 0.3 \cos(\theta) \text{ H}$$

$$L_{22} = 10 \cos(2\theta) + 30 \text{ H}$$

Using this information, please derive the torque $T_{fld}(\theta)$ for current $i_1 = 0.8 \text{ A}$ and $i_2 = 0.01 \text{ A}$

A17

The torque can be determined from Eq. (3.64).

$$T_{fld} = \frac{i_1^2}{2} \left(\frac{dL_{11}(\theta)}{d\theta} \right) + \frac{i_2^2}{2} \left(\frac{dL_{22}(\theta)}{d\theta} \right) + i_1 i_2 \left(\frac{dL_{12}(\theta)}{d\theta} \right).$$

For the given values of, $i_1 = 0.8 \text{ A}$ and $i_2 = 0.01 \text{ A}$, the torque (T_{fld}) is:

$$T_{fld} = -0.0024 \sin(\theta) - 0.00164 \sin(2\theta) \text{ N} \cdot \text{m}$$

As we can see in the equation, the torque expression consists of terms of two (2) types:

One term, proportional to $i_1 i_2 \sin \theta$, is due to the mutual interaction between the rotor and stator currents; it acts in a direction to align the rotor and stator so as to maximize their mutual inductance.

Alternately, it can be thought of as being due to the tendency of two magnetic fields (in this case those of the rotor and stator) to align.

The torque expression also has terms proportional to $\sin 2\theta$ and to the square of the individual coil currents. These terms are due to the action of the individual winding currents alone and correspond to the torques one sees in singly-excited systems. Here each torque component acts in a direction to maximize its corresponding inductance so as to maximize the coenergy. The 2θ torque variation is due to the 2θ variation in the self inductances, which in turn is caused by the variation of the air-gap reluctance; notice that rotating the rotor by 180° from any given position gives the same air-gap reluctance (hence the twice-angle variation). This torque component is known as the reluctance torque. The two (2) torque components (mutual and reluctance), along with the total torque, are plotted below.

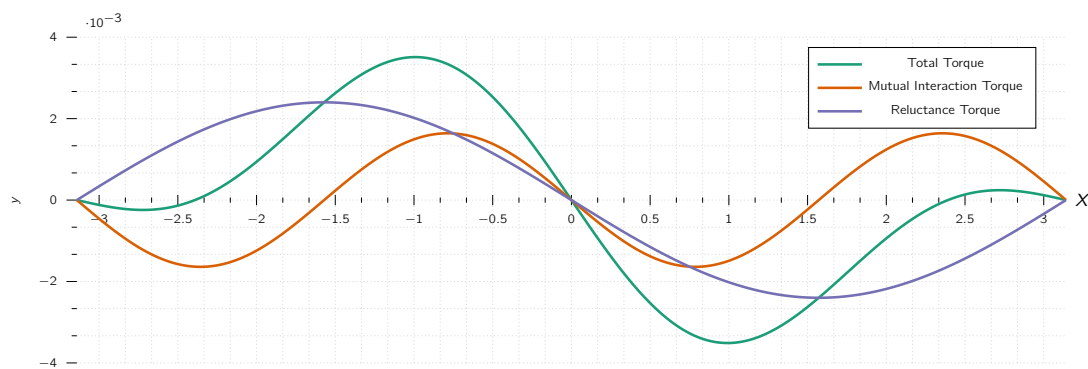


Figure 3.6: Plot of torque components for the multiply excited system of the exercise Multiply Excited System.

Rotating Magnetic Fields

Q18 Six-Phase Rotating Magnetic Field

We have seen a three-phase rotating magnetic field (RMF) and how it was generated. For this exercise, here is a six-phase current waveform input. Please calculate the generated magnetic flux density (B_{net}). The currents flowing through each phases (A, B, C, D, E, F) are as follows:

$$\begin{aligned} i_{AA'} &= I_{\max} \sin(wt), & i_{BB'} &= I_{\max} \sin(wt - 60^\circ), \\ i_{CC'} &= I_{\max} \sin(wt - 120^\circ), & i_{DD'} &= I_{\max} \sin(wt - 180^\circ), \\ i_{EE'} &= I_{\max} \sin(wt - 240^\circ), & i_{FF'} &= I_{\max} \sin(wt - 300^\circ). \end{aligned}$$

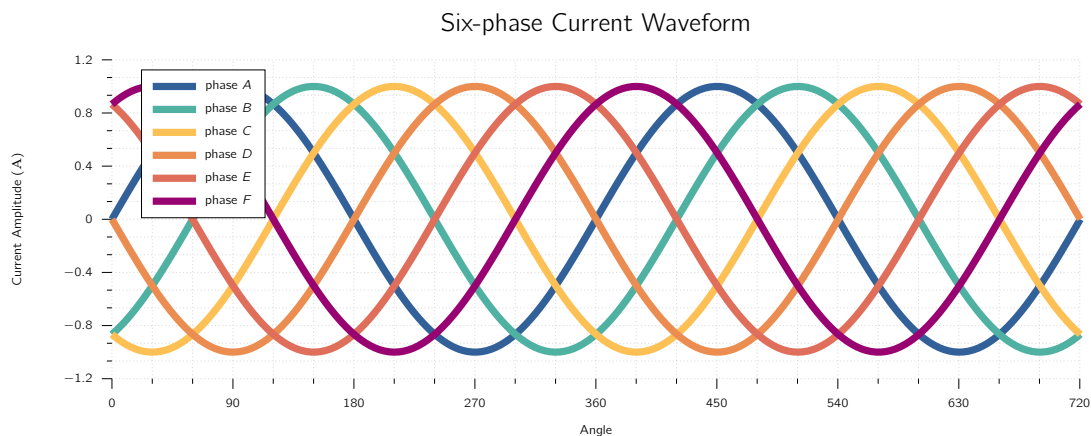


Figure 4.1: A graphic representation of a six-phase current waveform.

$$B_{\text{net}} = \left(\frac{2}{6}\right) B_{\max} [\sin(wt) - \cos(wt)]$$

Answer

A18

Let us first consider, what happens at the time instant of the voltage wave-forms. Unlike a three-phase system, at the instant $\theta = 90^\circ$, the voltage in phase **A** is positive maximum (1.0), phases **B** and **F** are half amplitude (0.5), **E** and **C** are also half amplitude but opposite sign (-0.5) and phase **D** is negative maximum (-1.0).

These six (6) wave-forms are represented by the following equations:

$$v_{AA'} = V_{\max} \sin(wt), \quad v_{BB'} = V_{\max} \sin(wt - 60^\circ),$$

$$\begin{aligned} v_{CC'} &= V_{\max} \sin(wt - 120^\circ), & v_{DD'} &= V_{\max} \sin(wt - 180^\circ), \\ v_{EE'} &= V_{\max} \sin(wt - 240^\circ), & v_{FF'} &= V_{\max} \sin(wt - 300^\circ). \end{aligned}$$

where $\omega = 2\pi \text{ rad} \cdot \text{s}^{-1}$, $V_{\max}$ is the maximum value of the voltage or induced EMF in each phase and f is the supply frequency (Hz).

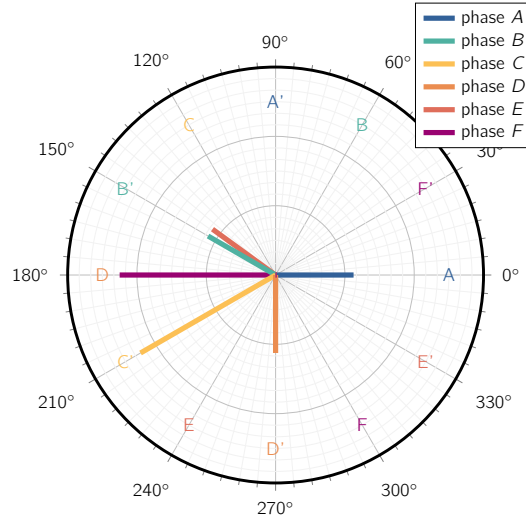


Figure 4.2: Polar coordinate form of the acting magnetic flux densities for the six phases at the electrical angle of $[e] = 30^\circ$.

If the magnetic flux densities (or fluxes (ϕ)) as both are valid approaches) are added to find the resultant magnetic flux density in phasor form the magnitude is found as;

$$\begin{aligned} B_{\text{net}} &= B_{AA'} \sin(wt) \angle 0^\circ + B_{BB'} \sin(wt - 60^\circ) \angle 60^\circ \\ &\quad + B_{CC'} \sin(wt - 120^\circ) \angle 120^\circ + B_{DD'} \sin(wt - 180^\circ) \angle 180^\circ \\ &\quad + B_{EE'} \sin(wt - 240^\circ) \angle 240^\circ + B_{FF'} \sin(wt - 300^\circ) \angle 300^\circ, \\ B_{\text{net}} &= B_{\max} \sin(wt) [\hat{x} \cos(0^\circ) + \hat{y} \sin(0)] \\ &\quad + B_{\max} \sin(wt - 60^\circ) [\hat{x} \cos(60^\circ) + \hat{y} \sin(60^\circ)] \\ &\quad + B_{\max} \sin(wt - 120^\circ) [\hat{x} \cos(120^\circ) + \hat{y} \sin(120^\circ)] \\ &\quad + B_{\max} \sin(wt - 180^\circ) [\hat{x} \cos(180^\circ) + \hat{y} \sin(180^\circ)] \\ &\quad + B_{\max} \sin(wt - 240^\circ) [\hat{x} \cos(240^\circ) + \hat{y} \sin(240^\circ)] \\ &\quad + B_{\max} \sin(wt - 300^\circ) [\hat{x} \cos(300^\circ) + \hat{y} \sin(300^\circ)] \end{aligned}$$

$$\begin{aligned}
B_{\text{net}} &= B_{\text{max}} \times \sin(wt) \times [1 \hat{x} + 0 \hat{y}] \\
&+ B_{\text{max}} \times \sin(wt - 60^\circ) \times \left[\frac{1}{2} \hat{x} + \frac{\sqrt{3}}{2} \hat{y} \right] \\
&+ B_{\text{max}} \times \sin(wt - 120^\circ) \times \left[-\frac{1}{2} \hat{x} + \frac{\sqrt{3}}{2} \hat{y} \right] \\
&+ B_{\text{max}} \times \sin(wt - 180^\circ) \times [-1 \hat{x} + 0 \hat{y}] \\
&+ B_{\text{max}} \times \sin(wt - 240^\circ) \times \left[-\frac{1}{2} \hat{x} - \frac{\sqrt{3}}{2} \hat{y} \right] \\
&+ B_{\text{max}} \times \sin(wt - 300^\circ) \times \left[\frac{1}{2} \hat{x} - \frac{\sqrt{3}}{2} \hat{y} \right]
\end{aligned}$$

Let's rearrange the equation by their **cardinal components**, namely $\hat{x}$ and $\hat{y}$.

$$\begin{aligned}
B_{\text{net}} &= \left[B_{\text{max}} \sin(wt) + \frac{1}{2} B_{\text{max}} \sin(wt - 60^\circ) \right. \\
&\quad - \frac{1}{2} B_{\text{max}} \sin(wt - 120^\circ) - B_{\text{max}} \sin(wt - 180^\circ) \\
&\quad \left. - \frac{1}{2} B_{\text{max}} \sin(wt - 240^\circ) + \frac{1}{2} B_{\text{max}} \sin(wt - 300^\circ) \right] \hat{x} \\
&\quad + \left[\frac{\sqrt{3}}{2} B_{\text{max}} \sin(wt - 60^\circ) + \frac{\sqrt{3}}{2} B_{\text{max}} \sin(wt - 120^\circ) \right. \\
&\quad \left. - \frac{\sqrt{3}}{2} B_{\text{max}} \sin(wt - 240^\circ) - \frac{\sqrt{3}}{2} B_{\text{max}} \sin(wt - 300^\circ) \right] \hat{y}
\end{aligned} \tag{4.3}$$

We need to do the following trigonometric conversions if we want to simplify the aforementioned equation of a mess

These conversions are based on the following trigonometric identities:

$$\begin{aligned}
\sin(x + 180^\circ) &= -\sin(x), \\
\sin(x + 360^\circ) &= +\sin(x).
\end{aligned}$$

$$\begin{aligned}
\sin(wt + 300^\circ + 360^\circ) &= +\sin(wt + 60^\circ), \\
\sin(wt - 240^\circ + 360^\circ) &= +\sin(wt + 120^\circ), \\
\sin(wt - 180^\circ + 180^\circ) &= -\sin(wt).
\end{aligned}$$

Inserting these equations to Eq. (4.3),

$$\begin{aligned}
B_{\text{net}} &= \left[B_{\text{max}} \sin(wt) + \frac{1}{2} B_{\text{max}} \sin(wt - 60^\circ) \right. \\
&\quad - \frac{1}{2} B_{\text{max}} \sin(wt - 120^\circ) + B_{\text{max}} \sin(wt) \\
&\quad \left. - \frac{1}{2} B_{\text{max}} \sin(wt + 120^\circ) + \frac{1}{2} B_{\text{max}} \sin(wt + 60^\circ) \right] \hat{x} \\
&\quad + \left[\frac{\sqrt{3}}{2} B_{\text{max}} \sin(wt - 60^\circ) + \frac{\sqrt{3}}{2} B_{\text{max}} \sin(wt - 120^\circ) \right. \\
&\quad \left. - \frac{\sqrt{3}}{2} B_{\text{max}} \sin(wt + 120^\circ) - \frac{\sqrt{3}}{2} B_{\text{max}} \sin(wt + 60^\circ) \right] \hat{y}
\end{aligned}$$

Now we can do some tidying up and do some arrangements of these sine values;

$$\begin{aligned} \mathbf{B}_{\text{net}} = & \left[2B_{\text{max}} \sin(\omega t) + \frac{1}{2}B_{\text{max}} \left(\sin(\omega t - 60^\circ) + \sin(\omega t + 60^\circ) \right) \right. \\ & \left. - \frac{1}{2}B_{\text{max}} \left(\sin(\omega t - 120^\circ) + \sin(\omega t + 120^\circ) \right) \right] \hat{x} \\ & + \left[-\frac{\sqrt{3}}{2}B_{\text{max}} \left(\sin(\omega t - 60^\circ) - \sin(\omega t + 60^\circ) \right) \right. \\ & \left. - \frac{\sqrt{3}}{2}B_{\text{max}} \left(\sin(\omega t + 120^\circ) - \sin(\omega t - 120^\circ) \right) \right] \hat{y} \end{aligned}$$

Now it is time to do some trigonometric manipulation;

These conversions are based on the following trigonometric identities:

$$\sin(A + B) + \sin(A - B) = 2 \sin(A) \cos(B),$$

$$\sin(A + B) - \sin(A - B) = 2 \cos(A) \sin(B).$$

and come up with our final equation to explain this rotating magnetic field for a six-phase system.

$$\begin{aligned} \mathbf{B}_{\text{net}} = & \left[2B_{\text{max}} \sin(\omega t) + \frac{1}{2}B_{\text{max}} \left(2 \sin(\omega t) \cos(60^\circ) \right) \right. \\ & \left. - \frac{1}{2}B_{\text{max}} \left(2 \sin(\omega t) \cos(120^\circ) \right) \right] \hat{x} \\ & + \left[-\frac{\sqrt{3}}{2}B_{\text{max}} \left(2 \cos(\omega t) \sin(60^\circ) \right) \right. \\ & \left. - \frac{\sqrt{3}}{2}B_{\text{max}} \left(2 \sin(\omega t) \cos(120^\circ) \right) \right] \hat{y} \\ \mathbf{B}_{\text{net}} = & B_{\text{max}} \left[2B_{\text{max}} \sin(\omega t) + 0.5 \sin(\omega t) + 0.5 \sin(\omega t) \right] \hat{x} \\ & + B_{\text{max}} \left[-\frac{3}{2} \cos(\omega t) - \frac{3}{2} \cos(\omega t) \right] \hat{y} \\ \mathbf{B}_{\text{net}} = & 3B_{\text{max}} \sin(\omega t) \hat{x} - 3B_{\text{max}} \cos(\omega t) \hat{y}, \\ \mathbf{B}_{\text{net}} = & \left(\frac{6}{2} \right) B_{\text{max}} \left[\sin(\omega t) \hat{x} - \cos(\omega t) \hat{y} \right] \quad \blacksquare \end{aligned}$$

Q19 Studying MMF Waves

A single armature winding phase of a two-pole machine can be considered to consist of 8 N_c -turn, full-pitch coils connected in series, with each slot containing two coils. There are a total of 24 armature slots, and thus each slot is separated by $360^\circ / 24 = 15^\circ$.

Define the angle θ_a as being measured from the magnetic axis of phase a such that the four (4) slots containing the coil sides labeled a are at $\theta_a = 67.5^\circ, 82.5^\circ, 97.5^\circ$, and, 112.5° .

The opposite sides of each coil are thus found in the slots found at $-112.5^\circ, -97.5^\circ, -82.5^\circ$ and -67.5° , respectively. Assume this winding to be carrying current i_a .

- Write an expression for the space-fundamental Magneto-motive Force (MMF) produced by the two coils whose sides are in the slots at $\theta_a = 112.5^\circ$ and -67.5° .

- b. Write an expression for the space-fundamental MMF produced by the two coils whose sides are in the slots at $\theta_a = 67,5^\circ$ and $-112,5^\circ$.
- c. Write an expression for the space-fundamental MMF of the complete armature winding.
- d. Determine the winding factor k_w for this distributed winding.

$$(\mathcal{F}_{ag1})_{22,5^\circ} = \frac{4}{\pi} \left(\frac{2N_c i_a}{2} \right) \cos(\theta_a - 22,5^\circ)$$

$$(\mathcal{F}_{ag1})_{-22,5^\circ} = \frac{4}{\pi} \left(\frac{2N_c i_a}{2} \right) \cos(\theta_a + 22,5^\circ)$$

$$(\mathcal{F}_{ag1})_{\text{total}} = 4.88 N_c i_a \cos \theta_a$$

Answer

A19

- a. Noting that the magnetic axis of this pair of coils is at

$$\theta_a = (112,5^\circ - 67,5^\circ) / 2 = 22,5^\circ$$

and that the total ampere-turns in each slot is equal to $2N_c i_a$, the MMF produced by this pair of coils can be found from analogy with Eq. (4.4) to be

$$(\mathcal{F}_{ag1})_{22,5^\circ} = \frac{4}{\pi} \left(\frac{2N_c i_a}{2} \right) \cos(\theta_a - 22,5^\circ)$$

- b. This pair of coils produces the same space-fundamental mmf as the pair of part (a) with the exception that this mmf is centered at $\theta_a = -22,5^\circ$. Thus

$$(\mathcal{F}_{ag1})_{-22,5^\circ} = \frac{4}{\pi} \left(\frac{2N_c i_a}{2} \right) \cos(\theta_a + 22,5^\circ)$$

- c. By analogy with parts (a) and (b), the total space-fundamental mmf can be written as

$$\begin{aligned} (\mathcal{F}_{ag1})_{\text{total}} &= (\mathcal{F}_{ag1})_{-22,5^\circ} + (\mathcal{F}_{ag1})_{-7,5^\circ} + (\mathcal{F}_{ag1})_{7,5^\circ} + (\mathcal{F}_{ag1})_{22,5^\circ} \\ &= \frac{4}{\pi} \left(\frac{2N_c}{2} \right) i_a \\ &\quad \times \left[\cos(\theta_a + 22,5^\circ) + \cos(\theta_a + 7,5^\circ) + \cos(\theta_a - 7,5^\circ) + \cos(\theta_a - 22,5^\circ) \right] \\ &= \frac{4}{\pi} \left(\frac{7.66N_c}{2} \right) i_a \cos \theta_a \\ &= 4.88 N_c i_a \cos \theta_a \end{aligned}$$

- d. Recognizing that, for this winding $N_{ph} = 8N_c$, the total MMF of part (c) can be rewritten as

$$(\mathcal{F}_{ag1})_{\text{total}} = \frac{4}{\pi} \left(\frac{0.958N_{ph}}{2} \right) i_a \cos \theta_a$$

Comparison with Eq. (4.6) shows that for this winding, the winding factor is $k_w = 0.958$.

Q20 Uniform Air Gap

A four-pole synchronous Alternating Current (AC) generator with a smooth air gap has a distributed rotor winding with 264 series turns, a winding factor of 0.935, and an air gap of length 0,7 mm. Assuming the MMF drop in the electrical steel to be negligible, find the rotor-winding current required to produce a peak, space-fundamental magnetic flux density of 1,6 T in the machine air gap.

11,4 A

Answer

A20

The space-fundamental air-gap magnetic field can be found from the space-fundamental component of the air-gap MMF by dividing by the air-gap length (g). Multiplying by the permeability of free space (μ_0) then gives the space-fundamental air-gap magnetic flux density. Therefore, from Eq. (4.9):

$$(B_{ag1})_{peak} = \frac{\mu_0 (\mathcal{F}_{ag1})_{peak}}{g} = \frac{4\mu_0}{\pi g} \left(\frac{k_r N_r}{\text{poles}} \right) I_r$$

Solving for I_r gives:

$$\begin{aligned} I_r &= \left(\frac{\pi g \times \text{poles}}{4\mu_0 k_r N_r} \right) (B_{ag1})_{peak} \\ &= \left(\frac{\pi \times 7 \times 10^{-4} \times 4}{4 \times 4\pi \times 10^{-7} \times 0,935 \times 264} \right) 1,6 \\ &= 11.4 \text{ A} \quad \blacksquare \end{aligned}$$

Q21 Calculating Angular Velocity

Consider a three-phase stator excited with balanced, 50 Hz currents. Find the synchronous angular velocity in rad/sec and speed in r/min for stators with two poles.

A21

For a frequency of $f_e = 50$ Hz, the electrical angular frequency is equal to:

$$\omega_e = 2\pi f_e = 100\pi \text{ rad} \cdot \text{s}^{-1}$$

Q22 Generated Voltage

A two-pole, three-phase, Y-connected, 60-Hz, round-rotor synchronous generator has a distributed field winding with N_f series turns and winding factor k_f . The armature winding has N_{ph} turns per phase and winding factor k_w .

The air-gap length is g , and the mean air-gap radius is r . The armature-winding active length is l . The dimensions and winding data are as follows:

$$N_f = 68 \quad \text{series turns}$$

$$k_f = 0.945$$

$$\begin{aligned}
 N_{\text{ph}} &= 18 \quad \text{series turns/phase} & k_w &= 0.933 \\
 r &= 0.53 \text{ m} & g &= 4.5 \text{ cm} \\
 l &= 3.8 \text{ m}
 \end{aligned}$$

The rotor is driven by a steam turbine at a speed of 3600 r/min. For a field current of $I_f = 720$ A dc, calculate the following:

- The peak fundamental mmf $(F_{\text{ag1}})_{\text{peak}}$ produced by the field winding,
- The peak fundamental flux density $(B_{\text{ag1}})_{\text{peak}}$ in the air gap,
- The fundamental flux per pole Φ_p , and
- the rms value of the open-circuit voltage generated in the armature.

$$\begin{aligned}
 25.7 \text{ kV} & & 3.31 \text{ Wb} \\
 0.821 \text{ T} & & 2.94 \times 10^4 \text{ A} \cdot \text{Turn}
 \end{aligned}$$

Answer

A22

- a. From Eq. (4.9):

$$\begin{aligned}
 (F_{\text{ag1}})_{\text{peak}} &= \frac{4}{\pi} \left(\frac{k_f N_f}{\text{poles}} \right) I_f = \frac{4}{\pi} \left(\frac{0.945 \times 68}{2} \right) 720 \\
 &= \frac{4}{\pi} (32.1) 720 = 2.94 \times 10^4 \text{ A} \cdot \text{Turn} \cdot \text{pole}^{-1}
 \end{aligned}$$

- b. Using Eq. (4.10), we get:

$$(B_{\text{ag1}})_{\text{peak}} = \frac{\mu_0 (F_{\text{ag1}})_{\text{peak}}}{g} = \frac{4\pi \times 10^{-7} \times 2.94 \times 10^4}{4.5 \times 10^{-2}} = 0.821 \text{ T}$$

Because of the effect of the slots containing the armature winding, most of the air-gap flux is confined to the stator teeth. The flux density in the teeth at a pole center is higher than the value calculated in part (b), probably by a factor of about 2. In a detailed design this flux density must be calculated to determine whether the teeth are excessively saturated.

- c. From Eq. (4.42):

$$\Phi_p = 2(B_{\text{ag1}})_{\text{peak}} l_r = 2(0.821)(3.8)(0.53) = 3.31 \text{ Wb}$$

- d. From Eq. (4.47) with $f_e = 60$ Hz

$$\begin{aligned}
 E_{\text{rms, line-neutral}} &= \sqrt{2} \pi f_e k_w N_{\text{ph}} \Phi_p = \sqrt{2} \pi (60)(0.933)(18)(3.31) \\
 &= 14.8 \text{ kV rms}
 \end{aligned}$$

The line-line voltage is thus

$$E_{\text{rms, line-line}} = \sqrt{3}(14.8 \text{ kV}) = 25.7 \text{ kV rms}$$

Q23

Rotor Torque

Consider a 4-pole, three-phase Synchronous Machine (SynM) with a uniform air gap. Assume the armature-winding self and mutual inductances to be constant:

$$L_{aa} = L_{bb} = L_{cc}$$

$$L_{ab} = L_{bc} = L_{ca}$$

Similarly, assume the field-winding self inductance L_f to be constant while the mutual inductances between the field winding and the three armature phase windings will vary with the angle θ_m between the magnetic axis of the field winding and that of phase a

$$\mathcal{L}_{af} = L_{af} \cos 2\theta_m$$

$$\mathcal{L}_{bf} = L_{af} \cos (2\theta_m - 120^\circ)$$

$$\mathcal{L}_{cf} = L_{af} \cos (2\theta_m + 120^\circ)$$

Show that when the field is excited with constant current I_f and the armature is excited by balanced-three-phase currents of the form:

$$i_a = I_a \cos (\omega_e t + \delta)$$

$$i_b = I_a \cos (\omega_e t - 120^\circ + \delta)$$

$$i_c = I_a \cos (\omega_e t + 120^\circ + \delta)$$

the torque will be constant when the rotor travels at synchronous speed ω_s .

$$\frac{\partial W'_{fid}}{\partial \theta_m} = 0$$

Answer

A23

The torque can be calculated from the coenergy as described in Section 3.6. This particular machine is a four-winding system and the coenergy will consist of four terms involving 1/2 the self-inductance multiplied by the square of the corresponding winding current as well as product-terms consisting of the mutual inductances between pairs of windings multiplied by the corresponding winding currents. Noting that only the terms involving the mutual inductances between the field winding and the three armature phase windings will contain terms that vary with θ_m , we can write the coenergy in the form:

The torque can now be found from the partial derivative of W'_{fid} with respect to θ_m

$$\begin{aligned} T &= \left. \frac{\partial W'_{fid}}{\partial \theta_m} \right|_{i_a, i_b, i_c, i_f} \\ &= -3L_{af}I_aI_f \sin(2\theta_m - \omega_e t - \delta) \end{aligned}$$

From this expression, we see that the torque will be constant when the rotor rotates at synchronous velocity ω , such that

$$\theta_m = \omega_s t = \left(\frac{2}{\text{poles}} \right) \omega_e t = \left(\frac{\omega_e}{2} \right) t$$

in which case the torque will be equal to

$$T = 3L_{af}I_aI_f \sin \delta$$

Note the torque for this three-phase machine operating at synchronous velocity under balanced-three-phase conditions is constant. As we have seen, this is due to the fact that the stator mmf wave consists of a single rotating flux wave, as opposed to the single-phase case in which the stator phase current produces both a forward- and a backward-rotating flux wave. This backwards flux wave is not in synchronism with the rotor and hence is responsible for the double-frequency time-varying torque component.

Q24**Estimating Maximum Torque**

An $1,800 \times 10^3$ rpm, four-pole, 60 Hz SynM has an air-gap length of 1,2 mm. The average diameter of the air-gap is 27 cm, and its axial length is 32 cm.

The rotor winding has 786 turns and a winding factor (k_w) of 0.976. Assuming that thermal considerations limit the rotor current to 18 A, please estimate the maximum torque and power output one can expect to obtain from this machine.

$$\omega \cdot N_e 0T \times 682'1$$

Answer

A24

First, we can determine the maximum rotor mmf from Eq. (4.9)

$$(F_r)_{\max} = \frac{4}{\pi} \left(\frac{k_r N_r}{\text{poles}} \right) (I_r)_{\max} = \frac{4}{\pi} \left(\frac{0.976 \times 786}{4} \right) 18 = 4395 \text{ A}$$

Assuming the peak value of the resultant air-gap flux is limited to 1,5 T, we can estimate the maximum torque from Eq. (4.74) by setting δ_r equal to $-\pi/2$ (recognising negative values of δ_r , with the rotor MMF lagging the resultant MMF, correspond to positive, motoring torque).

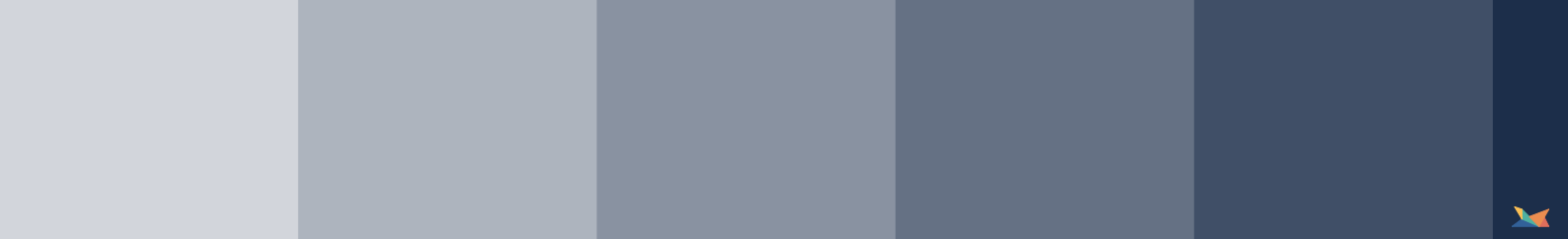
$$\begin{aligned} T_{\max} &= \left(\frac{\text{poles}}{2} \right) \left(\frac{\pi D l}{2} \right) B_{sr} (F_r)_{\max} \\ &= \left(\frac{4}{2} \right) \left(\frac{\pi \times 0.27 \times 0.32}{2} \right) 1.5 \times 4395 \\ &= 1,789 \times 10^3 \text{ N} \cdot \text{m} \end{aligned}$$

For a synchronous speed of $1,800 \times 10^3$ rpm:

$$\omega_m = n_s \left(\frac{\pi}{30} \right) = 1800 \left(\frac{\pi}{30} \right) = \frac{60}{\pi} \text{ rad} \cdot \text{s}^{-1}$$

and therefore the corresponding power can be calculated as:

$$P_{\max} = \omega_m T_{\max} = 337 \text{ kW} \quad \blacksquare$$



Glossary

AC An electric current that periodically reverses direction and changes its magnitude continuously with time, in contrast to direct current (DC), which flows only in one direction.. 25

MMF The magnetic pressure which drives magnetic flux in a magnetic circuit, and is the work done to move a unit magnetic pole.. 23–25

SynM Synchronous Machine. 27, 28